

Lending Club Loan Default Prediction using Deep Learning

Overview

This project builds an end-to-end loan default prediction system using TensorFlow-based Deep Learning. The solution analyzes Lending Club loan data and predicts whether a loan will be fully repaid or default.

Key Features

- Automated dataset profiling
- Feature engineering
- Correlation analysis
- Missing value analysis
- Categorical encoding
- Feature scaling
- SMOTE class balancing
- TensorFlow MLP model
- ROC-AUC evaluation
- Confusion matrix analysis
- Interactive HTML reporting

Dataset

Input dataset:

```
loan_data (2).csv
```

Target column:

```
not.fully.paid
```

Value	Meaning
0	Fully Paid
1	Not Fully Paid

Feature Engineering

Generated features:

- annual.inc
- installment_income_ratio

- revolbal_income_ratio

Data Preparation

- Missing value assessment
- Duplicate detection
- High correlation removal
- Label Encoding
- One-Hot Encoding
- Standard Scaling
- SMOTE Oversampling

Neural Network Architecture

```
Input Layer
↓
Dense(128) ReLU
↓
Dropout(0.25)
↓
Dense(64) ReLU
↓
Dropout(0.25)
↓
Dense(32) ReLU
↓
Dense(1) Sigmoid
```

Training Configuration

- Optimizer: Adam
- Learning Rate: 0.001
- Loss: Binary Crossentropy
- Epochs: 40
- Batch Size: 128
- Early Stopping Enabled
- Learning Rate Reduction Enabled

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix
- True Positives / Negatives
- False Positives / Negatives

Generated Visualizations

- Class Distribution
- Loan Purpose Distribution
- Default Rate by Purpose
- Interest Rate Distribution
- FICO Distribution
- Correlation Heatmap
- ROC Curve
- Training History
- Confusion Matrix

Output Report

```
reports/lending_club_dl_report.html
```

Run

```
python lending_club.py
```

Project Structure

```
project/  
├─ lending_club.py  
├─ loan_data (2).csv  
├─ reports/  
│   └─ lending_club_dl_report.html  
└─ README.md
```